

TWO-STAGE NONLINEAR MIXED MODEL ANALYSIS USING EXPECTED VALUE TRANSFORMATIONS

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Summary: The standard two-stage method is a simple procedure for estimating parameters of the population mean curve for the nonlinear random coefficient regression model, but it produces biased estimates with unknown distributions. This simple method can be improved by first reparameterizing the model using the expected value transformation. We propose an adaptation of the standard two-stage method in which the first step is to reparameterize the model, then apply the standard two-stage method to the expected value parameters rather than to the original parameters, and finally transform back to the original parameterization. The resulting estimates of the expected value parameters are approximately unbiased and normally distributed, consequently hypothesis tests and confidence intervals can be constructed using standard methods. Profile likelihood ellipses for the expected value parameters can be inverted to construct profile likelihood plots and confidence intervals for the original parameters. The final parameter estimates have smaller bias and variance than the standard two-stage estimates, and the confidence intervals are asymmetric, reflecting the actual distributions of parameter estimates more accurately. We

demonstrate fitting a Mitscherlich function to the effect of ambient CO_2 on photosynthesis of grasses (Potvin, Lechowicz, and Tardif 1990), comparing curves among treatment groups, and constructing confidence intervals. The coverage probabilities of confidence intervals for the standard two-stage method and the proposed method are compared in a Monte Carlo simulation.

1. INTRODUCTION

Repeated measures designs offer an efficient way to measure plant growth over time or to estimate effects of changing environmental conditions on plant responses such as photosynthesis. We focus on designed experiments in which the within-individual design matrices are similar for all individuals. The objective is to fit a nonlinear model across the levels of the repeated measures factors while taking into account the correlations among observations within an individual. Our interest centres in estimating the population mean curve and obtaining realistic confidence bands, rather than in predicting the response for any one particular individual, although the latter is not precluded. An example of this type of experiment appeared in Potvin, Lechowicz and Tardif (1990) which described a study of cold tolerance of a C_4 grass species. Twelve plants, divided into four treatment groups of three plants, were each subjected to a series of seven increasing ambient CO_2 concentrations. The treatments were factorial combinations of two chilling regimes and two ecological populations. Net photosynthesis was measured at each CO_2 level for each plant. A Mitscherlich function was fitted to each treatment group with the objective of estimating the effect of chilling on photosynthesis for each population and comparing the populations' tolerances to cold.

In recent years there has been a proliferation of papers dealing with alternative methods of analysis of repeated measures designs; Lindstrom and Bates (1990), Vonesh and Carter (1992), Davidian and Gallant (1993) and Wolfinger (1993), to mention a few. A bibliography is given by Davidian (1993). These papers look at ways of replacing the standard two-stage

estimation process by incorporating between and within individual stochastic modelling into one process. This is an elegant and commendable development but it does have some drawbacks. The primary drawback, in our opinion, is the need to specify some form of between individual distribution for the random parameters, ranging from the one extreme of Lindstrom and Bates who look at Normality to the other extreme of Davidian and Gallant, who only specify some degree of smoothness. In this accommodation, the authors require much more sophistication than the two-stage method, incorporating Bayesian, empirical Bayesian or semi-nonparametric smoothing. The properties of the resulting estimators still require study. To quote Davidian and Gallant, "We caution however, that validity, reliability and comparative performance of procedures based on empirical Bayes estimates are an open problem." In contrast, parameter transformation as recommended in this paper, together with two-stage least squares is easy to understand and implement.

It seems that a move away from an optimally transformed two-stage analysis to one combined approach need not necessarily be a good idea. The behaviour of estimators is not properly known and the need for specifying "between individual" distributions for random effects is a distinct drawback. Clearly (as pointed out by a referee), it is advantageous to be able to examine the distribution of random effects after stage one of the analysis and take advantage of this extra information wherever possible.

1.1 The random coefficient regression model

The nonlinear random coefficient regression (RCR) model is a natural and intuitively pleasing model for repeated measures data. In the RCR model each plant is assumed to have its own set of parameters, which are randomly drawn from a population of possible parameter vectors. In addition, the observations on a given plant fluctuate randomly around the plant's expected curve. The fact that each plant has its own expected curve induces a specific correlation structure on the observations within a plant. This model is appropriate when each plant is thought to respond to environmental

conditions with its own response curve, which is one realization from the universe of possible curves. It has the advantage that it can be applied when the number of individuals is small or when the design matrices are not exactly the same for all plants. The RCR model is

$$(1) \quad y_i = f(X_i, \theta_i) + \epsilon_i,$$

where $i = 1, \dots, n$ individuals. In this model y_i is the i -th individual's $t \times 1$ response vector, X_i is the within-individual design matrix, θ_i is the i -th individual's $p \times 1$ parameter vector, and ϵ_i is a vector of within-individual errors, where there are t measurements on each individual and p parameters in the model for the mean. The function f has a known form, and in this paper we specifically concentrate on the function f being non-linear in θ . For example, in the cold tolerance experiment for one treatment $n = 3$, $t = 7$, $p = 3$ parameters of the Mitscherlich function, and y_i is a 7×1 vector containing the response values for the i -th plant. The parameter vector θ_i is assumed to follow some probability distribution with mean vector θ and covariance matrix Σ_θ which is a $p \times p$ matrix. In this example X_i is just a 7×1 vector of ambient CO_2 concentrations. Each plant could be measured at different CO_2 concentrations and the number of measurements need not be the same for all plants.

In the RCR model, random coefficients are embedded in the model for the mean. As a result the distribution of y is not simple when the model for the mean is a nonlinear function. Contrasting the nonlinear RCR model with the linear RCR model, we see that in the linear case, where $y_i = X_i \beta_i + \epsilon_i$, $\beta_i \sim N(\beta, \Sigma_\beta)$, and $\epsilon_i \sim N(0, \sigma^2 I)$, the distribution of y_i is exactly normal and its mean vector and covariance matrix can be written down explicitly. In the nonlinear random coefficient model, however, normality of θ_i does not imply that y_i is normally distributed. In general the distribution, mean vector, and covariance matrix of y_i are not known exactly.

In the case of the linear model with Gaussian errors, if the covariance matrix Σ_β is known, the generalized least squares (GLS) estimator $\hat{\beta}$ is unbiased and exactly normally distributed and confidence intervals for β can be constructed in a straightforward way. The mean response curve is

estimated as $X\hat{\beta}$, which is also exactly normally distributed. If the design matrices are the same for all individuals, then the generalized least squares estimator is the same as the standard two-stage estimator $\bar{b}$, the average of the n individual estimated coefficient vectors $b_i = (X'X)^{-1}X'y_i$ (Swamy 1971).

When the function f is nonlinear in θ , however, the standard two-stage estimator $\bar{b}$ no longer coincides with the GLS estimator even when the design matrices are the same for all individuals. Estimation methods for the nonlinear RCR model up to 1985 are summarized in Steimer, Mallet, and Mentre (1985). Beal (1984), Racine-Poon (1985), Berkey and Laird (1986), Lindstrom and Bates (1990), and Vonesh and Carter (1992) presented maximum likelihood, Bayesian, and estimated generalized least squares estimators for nonlinear mixed models, including random coefficient regression models.

1.2 Parameter transformations

For the nonlinear RCR model the estimator of θ is not normally distributed and indeed most often these distributions may be severely skewed and biased. The resulting problems of significance tests for θ and confidence regions can be tackled in a similar way to that for standard nonlinear regression. Perhaps the most valuable tool is parameter transformation (Bates and Watts 1981). By reparameterization, a nonlinear model may be made to behave more like a linear model in some transformed parameters ϕ within the range of the data. The expected value transformation (Ross 1975, Ratkowsky 1990), which uses the parameter vector $\phi = f(u, \theta)$ rather than the original parameter vector θ , may be made to achieve this objective to a remarkably high degree of exactitude. Technically it is explained by a reduction in parameter-effects curvature (Bates and Watts 1980, Clarke 1987).

It is possible to find values of the regressor variables u for which the likelihood contours of $\hat{\phi}$ are nearly elliptical or for which $\hat{\phi}$ is unbiased (Clarke 1987). We apply the expected value transformation to improve

point estimates of the parameters and to improve the coverage probability of confidence intervals for the nonlinear random coefficient regression model.

We propose adaptation of the standard two-stage or $\bar{b}$ method (Steimer et al. 1985; Gumpertz and Pantula 1989), in which a separate curve is fit to each plant and population mean parameter estimates are obtained by averaging parameter estimates over all plants. The first step in the proposed adaptation is to reparameterize the model using an expected value transformation. After reparameterization, the procedure follows the standard two-stage method, except that expected value parameter estimates, $\hat{\phi}$, are averaged rather than parameters of the original model. When y_i is normally distributed, and a separate model is fit for each individual, the distributions of the ordinary least squares estimators of ϕ are approximately Gaussian and have negligible bias. Consequently, standard methods of constructing confidence intervals and hypothesis tests for the mean response curves can be used. In particular, classical linear theory multivariate tests are possible. A direct extension is to transform back from confidence regions for ϕ to those for θ .

In section 2 we give details for implementing the parameter transformations for the RCR model, multivariate significance tests, and construction of profile confidence intervals for θ . An example is presented in section 3 illustrating how to fit the nonlinear RCR model, compare mean curves across treatments, construct confidence bands for the expected response curves, and construct profile likelihood plots or confidence curves (Cook and Weisberg 1990) for the θ parameters. A simulation is presented in section 4, and our conclusions are summarized in section 5.

2. TRANSFORMATION UNDER THE RANDOM COEFFICIENT REGRESSION MODEL

For a given individual, assume that the random coefficient regression model (1) holds. The model for the mean can be reparameterized to $f(X_i, \theta_i) = g(X_i, \phi_i)$, where ϕ_i is a $p \times 1$ vector of expected value parameters. The expected value parameters, $\phi_i = f(u, \theta_i)$, are just the values the mean

function takes at some specified points $\mathbf{u}$, which are called "transform points". For example, the model $f(\mathbf{X}, \boldsymbol{\theta}) = \log(\alpha + \beta\mathbf{X})$, where $\boldsymbol{\theta}' = [\alpha \ \beta]$, can be re-expressed as

$$g(\mathbf{X}, \boldsymbol{\phi}) = \log \left\{ \frac{X_2 - X}{X_2 - X_1} e^{\phi_1} + \frac{X - X_1}{X_2 - X_1} e^{\phi_2} \right\},$$

where ϕ_1 and ϕ_2 are the expected values of Y at X_1 and X_2 , respectively, X_1 and X_2 are any two values in the range of the data, and $\boldsymbol{\phi}' = [\phi_1 \ \phi_2]$ (Ratkowsky 1990). Not all models can be re-expressed in such a tidy form; nevertheless, given a model, f , with p parameters we can always find p expected values to correspond to p levels of X . When an exact inverse transformation is not explicitly available, we can convert from $\boldsymbol{\phi}$ to $\boldsymbol{\theta}$ iteratively by fitting the nonlinear regression model $\boldsymbol{\theta} = f(\mathbf{u}, \boldsymbol{\theta})$.

In the proposed method, first fit a separate curve by ordinary nonlinear least squares (e.g. using SAS[®] PROC NLIN) for each of the n individuals in the sample using the original model, $y_i = f(\mathbf{X}_i, \boldsymbol{\theta}_i) + \epsilon_i$, $i = 1, \dots, n$. Denote the resulting parameter estimates $\hat{\boldsymbol{\theta}}_i$. Then choose a vector $\mathbf{u}$ of p transform points. Estimate the i -th individual's expected values at the transform points by $\hat{\boldsymbol{\phi}}_i(\mathbf{u}) = f(\mathbf{u}, \hat{\boldsymbol{\theta}}_i)$, which is simply the vector of predicted values from the i -th regression. We assume that the within-individual errors, ϵ_i , are normally distributed and independent of $\boldsymbol{\phi}_i$. Application of this method does not require that ϵ_i be independent and identically distributed; the method should be equally applicable to any error structure, such as AR1. However, for simplicity, the detailed formulae given in this section assume $\text{Var}(\epsilon_i) = \sigma^2 \mathbf{I}$.

2.1 Choice of optimal transform points

The choice of $\mathbf{u}$ is a critical stage of the process. As we vary $\mathbf{u}$ so will we generate parameter estimates $\hat{\boldsymbol{\phi}} = f(\mathbf{u}, \hat{\boldsymbol{\theta}})$ whose distributional properties will vary. The ideal would be to find some $\mathbf{u}$ such that $\hat{\boldsymbol{\phi}}$ are exactly normally distributed. This ideal is unattainable but we can get close to it. It so happens that the bias in $\hat{\boldsymbol{\phi}}$ is a good indicator of the distributional properties and consequently, choice of $\mathbf{u}$ to give zero bias in $\hat{\boldsymbol{\phi}}$ results in

nearly normal distributions. The other advantage of using the zero bias criterion is that it is relatively easy to implement in practice. A full discussion of this and other transformation criteria is given in Clarke (1987).

The bias of the parameter vector can be estimated by substituting the estimates $\hat{\phi}_i$ into the quadratic approximation for bias due to Box (1971):

$$(2) \text{ Bias } (\hat{\theta}_i) = -\frac{1}{2} \text{Var}(\hat{\theta}_i) \frac{\partial f'(X_i, \theta_i)}{\partial \theta_i} [\text{Var}(y_i | \theta_i)]^{-1} \begin{bmatrix} \text{trace} \left[\text{Var}(\hat{\theta}_i) \left[\frac{\partial^2 f(X_{i1}, \theta_i)}{\partial \theta_i \partial \theta_i'} \right] \right] \\ \vdots \\ \text{trace} \left[\text{Var}(\hat{\theta}_i) \left[\frac{\partial^2 f(X_{ip}, \theta_i)}{\partial \theta_i \partial \theta_i'} \right] \right] \end{bmatrix}$$

$$\text{where } \text{Var}(\hat{\theta}_i) = \sigma_i^2 \left[\frac{\partial f'(X_i, \theta_i)}{\partial \theta_i} \frac{\partial f(X_i, \theta_i)}{\partial \theta_i'} \right]^{-1} \text{ and } \text{Var}(y_i | \theta_i) = \hat{\sigma}_i^2 = \text{MSE}$$

from the nonlinear regression for the i -th individual. The bias for a set of new parameters ϕ , which are functions of the original parameters θ , can be obtained from the bias of $\hat{\theta}_i$. If we start with the new parameterization, so that the model is expressed as $y_i = g(X_i, \phi_i)$, expression (2) yields the bias of $\hat{\phi}_i$ involving partial derivatives of g with respect to ϕ . These may be converted to expressions involving derivatives of θ using the chain rule for vectors (see, e.g., Seber and Wild 1989), which, when substituted into expression (2), results in

$$(3) \quad \text{Bias}(\hat{\phi}_i) = \left[\frac{\partial \phi_i}{\partial \theta_i'} \right] (\text{bias}(\hat{\theta}_i)) + \frac{1}{2} \begin{bmatrix} \text{trace} \left[\text{Var}(\hat{\theta}_i) \left[\frac{\partial^2 \phi_{1i}}{\partial \theta_i \partial \theta_i'} \right] \right] \\ \vdots \\ \text{trace} \left[\text{Var}(\hat{\theta}_i) \left[\frac{\partial^2 \phi_{pi}}{\partial \theta_i \partial \theta_i'} \right] \right] \end{bmatrix}$$

where $\left[\frac{\partial \phi_i}{\partial \theta_i'} \right]$ and $\left[\frac{\partial^2 \phi_{ri}}{\partial \theta_i \partial \theta_i'} \right]$, $r = 1, \dots, p$, are the first-order Jacobean and

second-order Hessian derivative matrices respectively, evaluated at $\theta = \hat{\theta}$.

To implement equation (3) in practice, all that is needed is to choose a succession of values of u to cover the range of observed X and evaluate the biases for each. Since the bias function changes smoothly with increasing/decreasing u , zero points may be immediately identified from sign changes in the bias.

The transform points for the i -th individual are the p levels of X that minimize this bias. If the ranges of X values are similar for all individuals, the transform points would be expected to be similar also. We suggest using the mean or median of the vectors of transform points across individuals to obtain a common set of transform points for all individuals, for example $u = \frac{1}{n} \sum_{i=1}^n u_i$, where u_i contains the p levels of X chosen for the i -th individual.

2.2 Tests to compare curves across treatments

Under the assumptions that $\epsilon_i \sim N(0, \sigma^2 I)$, ϕ_i is independent of ϵ_i and ϕ_i is independent of ϕ_j for all i and j , the vectors $\hat{\phi}_i(u)$ are independent and nearly normally distributed with $\hat{\phi}_i | \phi_i \sim N(\phi_i, \text{Var}(\hat{\phi}_i | \phi_i))$, where

$$\text{Var}(\hat{\phi}_i | \phi_i) \doteq \sigma^2 \left[\frac{\partial g(u, \phi_i)}{\partial \phi_i'} \frac{\partial g(u, \phi_i)}{\partial \phi_i} \right]^{-1}.$$

If we also assume between-individual normal distributions, that is $\phi_i \sim N(\phi, \Sigma_\phi)$, then the sample mean of the estimates,

$$\bar{\phi}(u) = \frac{1}{n} \sum_{i=1}^n \hat{\phi}_i(u),$$

has little bias and is approximately normally distributed with expectation $\phi(u)$ and variance

$$\text{Var}(\bar{\phi}(u)) \doteq \frac{1}{n} \Sigma_\phi = \frac{\sigma^2}{n^2} \sum_{i=1}^n \left[\frac{\partial g(u, \phi_i)}{\partial \phi_i'} \frac{\partial g(u, \phi_i)}{\partial \phi_i} \right]^{-1},$$

which is estimated by the sample covariance matrix of the $\hat{\phi}_i$ values,

$$S_{\phi} = \frac{1}{n-1} \sum_{i=1}^n (\hat{\phi}_i(u) - \bar{\phi}(u))(\hat{\phi}_i(u) - \bar{\phi}(u))'$$

(Gumpertz and Pantula 1989). T^2 confidence bands for the expected population curve are constructed by computing the following confidence limits for each of the $k = 1, \dots, p$ values in u :

$$\bar{\phi}_k \pm \sqrt{\frac{df \cdot p}{n(df-p+1)} F_{1-\alpha}(p, df-p+1) S_{\phi kk}},$$

where df = degrees of freedom for computing S_{ϕ} and n = number of individual curves. These are simultaneous confidence bands for all levels of X within the range covered by the experimental design.

If individuals have been assigned to different treatment groups and the objective is to estimate mean curves for each treatment, separate confidence bands can be constructed for the groups. Mean response curves may be compared using multivariate test statistics such as Wilks' lambda on the vectors of expected values for the different groups, $\bar{\phi}_h$, where $h = 1, \dots, g$ treatment groups. The pooled covariance matrix is

$$S_{\phi} = \frac{1}{n-g} \sum_{h=1}^g \sum_{i=1}^{n_h} (\hat{\phi}_{hi}(u) - \bar{\phi}_h(u))(\hat{\phi}_{hi}(u) - \bar{\phi}_h(u))',$$

where there are n_h individuals in the h -th treatment group and n_T individuals altogether. Since we are effectively analyzing the vector of ϕ parameters as if it were a linear model, the actual performing of standard multivariate tests can be done in a straightforward way using any statistical package (e.g. SAS[®] PROC GLM). There is the additional advantage that individual comparisons among the elements of $\hat{\phi}$ may yield useful and meaningful insight into reasons for significant differences between curves.

2.3 Back transformations to the θ parameters

Confidence curves can be obtained for the parameters of the original model or for functions of the parameters using the profile likelihood approach described in Clarke (1987). The principle is that for any specified level, α , we can determine a $(1-\alpha) \cdot 100\%$ profile likelihood ellipsoid in ϕ space, as all

values of ϕ that satisfy

$$(4) \quad n(\phi - \bar{\phi})' S_{\phi}^{-1} (\phi - \bar{\phi}) \leq F_{1-\alpha}(1, df),$$

where df indicates error degrees of freedom. Then, viewing the original parameters, θ_k , $k = 1, \dots, p$, as functions of the vector of expected value parameters, ϕ , there exists a minimum value of θ_k (and a maximum value of θ_k) whose corresponding ϕ lies in the ellipsoid. Then, approximately, we interpret the minimum value of θ_k as the $100(\frac{\alpha}{2})$ percentile and the maximum value of θ_k as the $100(1 - \frac{\alpha}{2})$ percentile of the profile likelihood limits to θ_k . Note that the use of one degree of freedom in $F_{1-\alpha}(1, df)$ corresponds to the confidence curve method of Cook and Weisberg (1990) and the profile- t method of Bates and Watts (1988). Clarke (1987) derived an expression for the ϕ vector corresponding to the smallest (or largest) value of θ_k that is in the ellipsoid (4). The vectors ϕ corresponding to the $100(\frac{\alpha}{2})$ and the $100(1 - \frac{\alpha}{2})$ percentiles of the profile likelihood for the random coefficient regression model are

$$(5) \quad \bar{\phi} \pm \frac{t_{\alpha/2}(df)}{\sqrt{n \left[\left[\frac{\partial \phi}{\partial \theta'} \right]^{-1} S_{\phi} \left[\frac{\partial \phi}{\partial \theta'} \right]^{-1} \right]_{kk}}} S_{\phi} \left[\frac{\partial \phi}{\partial \theta'} \right]^{-1} e_k,$$

where $e_k = [0 \ 0 \ \dots \ 1 \ \dots \ 0]'$ with the one in the k -th position. These are similar to Wald confidence intervals for ϕ except that the usual variance of $\bar{\phi}$ is replaced by $\sum_{l=1}^p \frac{\partial \phi}{\partial \theta_l} \text{Cov}(\theta_{kL}, \theta_{lL}) / \sqrt{\text{Var}(\theta_{kL})}$, where the subscript L indicates the estimated confidence limit.

The algorithm for finding approximate profile confidence limits for one parameter, θ_k , is as follows:

- (1) Let the subscripts "curr" and "new", respectively, denote the current and new estimates of θ and ϕ . At the very first cycle of the iteration set $\theta_{\text{curr}} = \bar{\theta}$ corresponding to $\phi_{\text{curr}} = \bar{\phi}$. Then

$$\phi_{\text{new}} = \bar{\phi} \pm \frac{t_{\alpha}(df)}{\sqrt{n \left[\left(\frac{\partial \phi'}{\partial \theta} \right)^{-1}_{\text{curr}} S_{\phi} \left(\frac{\partial \phi'}{\partial \theta} \right)^{-1}_{\text{curr}} \right]_{kk}}} S_{\phi} \left(\frac{\partial \phi'}{\partial \theta} \right)^{-1}_{\text{curr}} e_k,$$

$$\text{where } \frac{\partial \phi'}{\partial \phi_{\text{curr}}} = \frac{\partial \phi'}{\partial \theta} \bigg|_{\theta = \theta_{\text{curr}}}, \quad df = n - g.$$

- (2) Use the Gauss-Newton algorithm to find θ_{new} corresponding to ϕ_{new} . This is an iterative procedure, but only requires a few iterations since we are fitting p parameters to p observations.
- (3) Iterate steps (1) and (2) until the difference between θ_{new} and θ_{curr} is small. The k -th element of θ_{new} is the desired quantile of the profile likelihood for θ_k .
- (4) Repeat steps (1) through (3) for each of the parameters θ_k , $k=1, \dots, p$.

The great benefit of setting confidence limits to θ in this way is that the possible asymmetry in these limits reflects our uncertainty in θ , as measured by changes in the likelihood. It is not logical to define a confidence interval such that some points outside this interval have greater likelihood than others inside it.

3. EXAMPLE: COLD TOLERANCE OF A C_4 GRASS SPECIES

3.1 Description of the experiment

Potvin et al. (1990) studied the effect of chilling on photosynthesis for plant populations from two different climate regions. The data are shown in figure 1. Six plants from Mississippi seed and six from Quebec seed were grown under controlled conditions. After four weeks of growth, half of the plants from each population (Mississippi and Quebec) were subjected to one full day of chilling at 7°C . The remaining plants were kept at 26°C , the standard daytime temperature. Following one night of recovery, carbon uptake was measured on each plant at a series of seven ambient CO_2 concentrations.

The objective of the study was to compare carbon uptake curves for the two plant populations and to determine whether the effect of temperature was similar for the two populations. A Mitscherlich equation was fit to the response curve,

$$(6) \quad E(Y) = A(1 - e^{-k(X-X_0)})$$

The parameters have the following biological interpretations: A = saturated rate of CO_2 uptake; k = rate of increase in photosynthesis with enhanced CO_2 ; and X_0 = CO_2 compensation point. Potvin et al. (1990) pointed out that estimates of the same quantities can be obtained by fitting a rectangular hyperbola with the following parameterization:

$$E\left(\frac{1}{Y}\right) = \frac{1}{A\left(1 + \frac{1}{-k(X-X_0)}\right)},$$

but that the parameter estimates varied considerably for the two models. They state: "On the basis of the residual SS, we were unable to discriminate between the two models. However, estimated values of the parameters were very different, and could easily lead to biologically different interpretations. In our case, the values of the parameters were highly dependent upon the model chosen; this illustrates the potential danger in restricting the analysis to the coefficients, and emphasizes the advantage of comparing the overall curves." We concur in the conclusion that comparisons of the mean curves as a whole, rather than comparisons of the parameters of the model (6) are more informative for these data. The parameter X_0 , the point at which CO_2 uptake becomes positive, is outside the range of the data (fig. 1), so it is not surprising that different models may fit well over the range of the data but may have different values of X_0 . For the same reason the rate of increase in photosynthesis at X_0 , k , may differ for two models that appear to fit equally well.

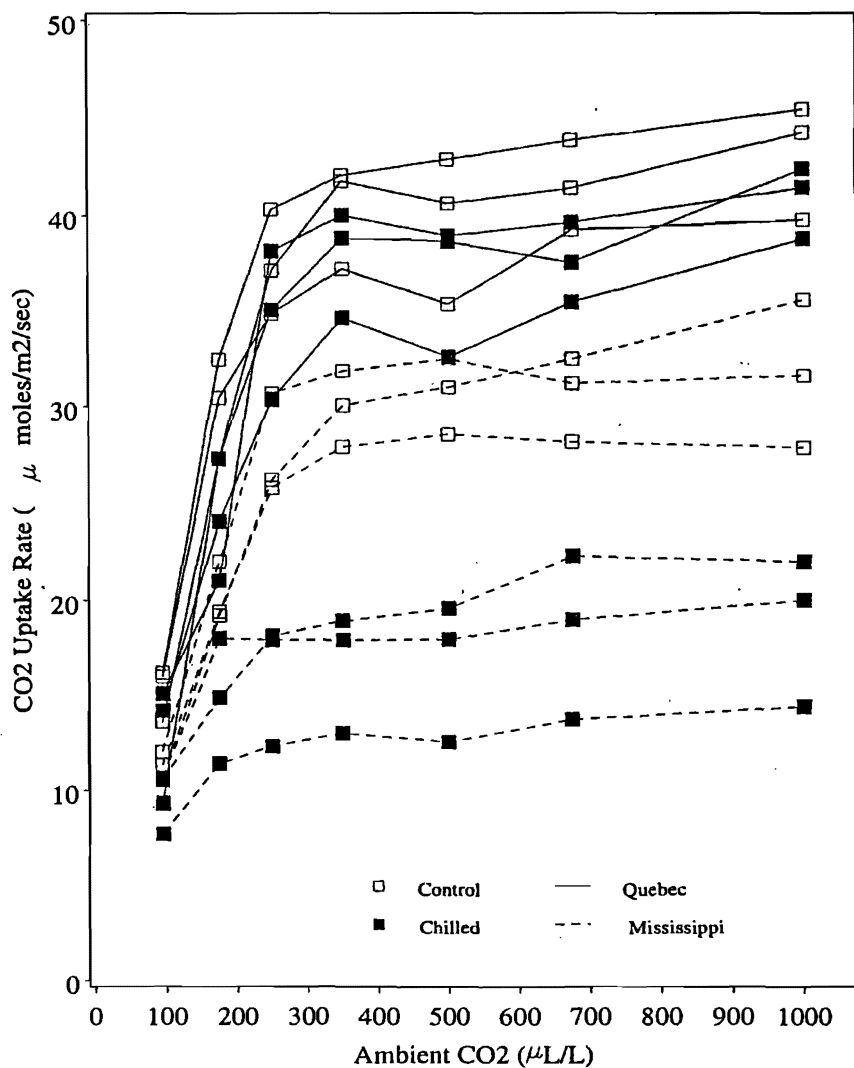


Figure 1. CO2 uptake curves (Porvin et al. 1990).

3.2 Random coefficient regression model

We first fitted a separate curve (equation 6) for each plant to obtain $\hat{\theta}_{hi}$ for $h = 1, \dots, 4$ treatment groups and $i = 1, \dots, 3$ plants per group, then computed the standard two-stage estimate, $\bar{\theta}_h$, for each group (table 1). In this dataset the Mississippi chilled treatment group is difficult to fit with the Mitscherlich function because the response for that group is almost flat (fig. 1), leading to extremely large estimated biases for $\bar{k}$ and $\bar{X}_0$ (table 1).

The estimated bias of $\bar{\theta}_h$ is $\frac{1}{n} \sum_{i=1}^3 \text{bias}(\hat{\theta}_{hi})$, where the bias vector for the

i -th plant, $\text{bias}(\hat{\theta}_{hi})$, is computed from equation (2) with MSE equal to the pooled residual variance from the 12 separate regressions. Note that the assumption of homogeneous variances may not be a good one for these data. Standard errors of the regression coefficients based on only the three plants in each group are shown in parentheses in table 1. For all three parameters the variability of the Mississippi chilled plants is larger than that of any other group. Since there were only three plants per group, however, the pooled variance, MSE, was used in all further analysis.

TABLE 1.

Standard two-stage estimates, $\bar{\theta}_h$, and estimated bias of $\bar{\theta}_h$ expressed as % of $\bar{\theta}_h$

Group	Parameter Estimates ^a			Estimated Bias (%)		
	A	k	X_0	A	k	X_0
QU Control	41.75(1.9)	0.0111(0.0009)	53.91(1.4)	0.13	2.1	1.8
QU Chilled	39.15(1.4)	0.0092(0.0053)	49.86(12.0)	0.18	1.6	2.6
MS Control	31.56(1.4)	0.0093(0.0054)	46.47(5.7)	0.29	3.1	4.3
MS Chilled	17.96(2.4)	0.016 (0.0092)	20.17(25.7)	1.23	132	58
Pooled						
Std Deviation	1.860	0.00406	14.48			

^a Standard errors based on estimates for 3 plants are given in parentheses after parameter estimates.

For each plant a vector of three minimum bias points, u_{hi} , was constructed using equation (3). The average of the u_{hi} vectors over all plants was chosen to be the vector of transform points, $u = [100, 233, 588]'$. Next, the predicted values at the transform points, $\hat{\phi}_{hi}(u)$, were computed for each plant and averaged over the three plants in each treatment group (table 2). The minimum bias points for the twelve plants ranged from 97 to 106 for u_1 , 211 to 279 for u_2 , and 340 to 746 for u_3 . In spite of the large range for u_3 the estimated bias of $\bar{\phi}$ was less than 6% of $\bar{\phi}$ for all 4 groups and all 3 transform points (table 2). For even the worst possible choice of transform points, the estimated bias of $\bar{\phi}$ was less than 10% (fig. 2).

TABLE 2

$\bar{\phi}_h$ at $u_1 = 100$, $u_2 = 233$, and $u_3 = 588$ and estimated bias of $\bar{\phi}_h$ expressed as % of $\bar{\phi}_h$

Group	$\bar{\phi}$			Estimated Bias (%)		
	100	233	588	100	233	588
QU Control	16.56	35.77	41.60	0.08	0.05	0.02
QU Chilled	13.56	31.56	38.79	0.00	0.02	0.03
MS Control	11.95	25.44	31.15	0.03	0.00	0.03
MS Chilled	10.28	15.82	17.74	5.97	1.00	0.03

The pooled covariance matrix of the $\hat{\phi}_{hi}(u)$ values is

$$S_{\phi} = \begin{bmatrix} 3.108 & 1.289 & 0.7440 \\ 1.289 & 6.535 & 5.802 \\ 0.7440 & 5.802 & 9.332 \end{bmatrix}.$$

Multivariate repeated measures analysis of the $\hat{\phi}_{hi}(u)$ values provided tests for population, temperature, and ambient CO_2 main effects (main effects are averaged over levels of the remaining factors) and interactions (table 3). These tests appeared to be more powerful than the corresponding tests applied to the raw data (cf. Potvin et al. 1990, table 6) for all hypotheses involving contrasts among CO_2 levels. Separate analyses of variance of the $\hat{\phi}_j$ values indicated that the Population $\times$ Temperature

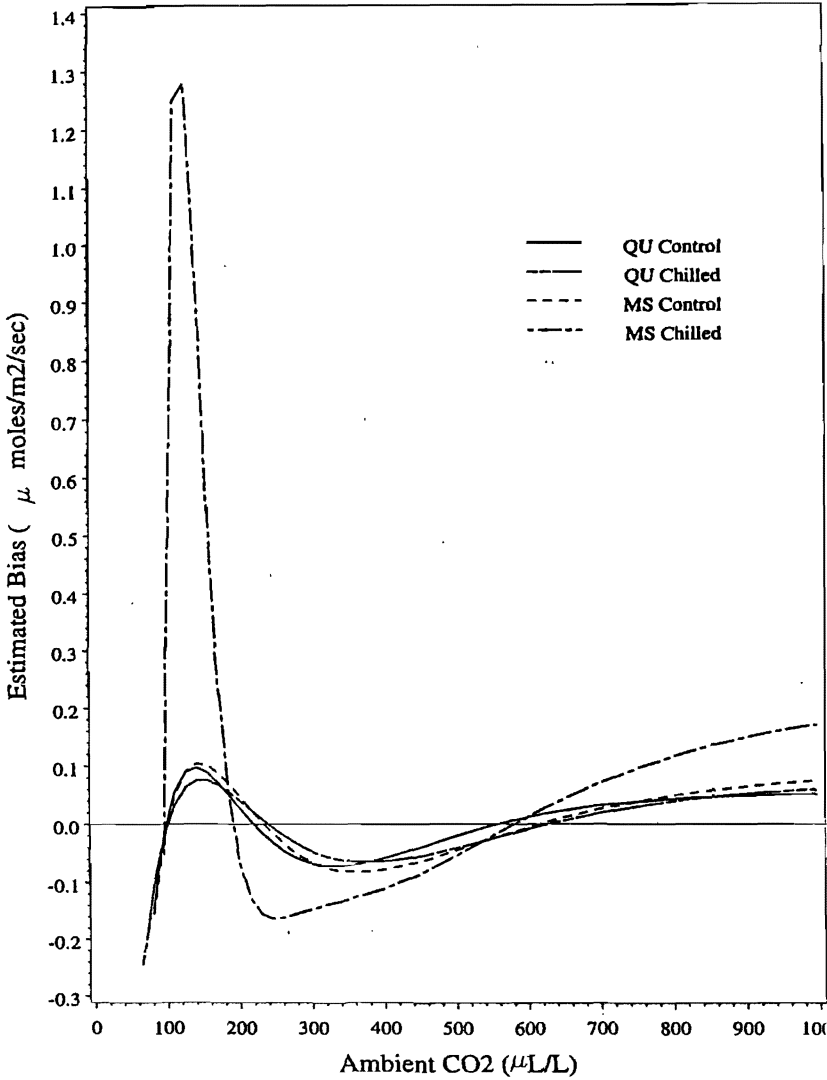


Figure 2. Estimated bias of $\bar{\varphi}$ for two plant populations and two chilling regimes

interaction was highly significant for ϕ_3 (p-value = 0.017), less so for ϕ_2 , and not at all significant for ϕ_1 .

TABLE 3
Multivariate repeated measures analysis of $\hat{\phi}_{hi}(u)$

Source	Wilks' λ	F	df	P-value
Population (M vs Q)		93	1, 8	0.0001
Temperature		26	1, 8	0.0009
P $\times$ T		4.7	1, 8	0.063
Ambient CO ₂	0.01859	185	2, 7	0.0001
P $\times$ Ambient CO ₂	0.1624	18	2, 7	0.0017
T $\times$ Ambient CO ₂	0.4417	4.4	2, 7	0.057
P $\times$ T $\times$ Ambient CO ₂	0.4464	4.3	2, 7	0.059

We also tested the following specific contrasts among the mean vectors of expected responses for different groups: (1) $\phi_{Q,control} = \phi_{Q,chilled}$; (2) $\frac{1}{2}(\phi_{Q,control} + \phi_{Q,chilled}) = \phi_{M,chilled}$; and (3) $\frac{1}{2}(\phi_{Q,control} + \phi_{Q,chilled}) = \phi_{M,control}$. The results were similar to those obtained by Potvin et al. (1990, table 8), indicating large differences between both Mississippi response curves and the average Quebec curve (Wilks' λ , p-value = 0.0063 and 0.0015 for chilled and control plants respectively) and little difference between the two Quebec curves (Wilks' λ , p-value = 0.27).

The mean expected values were converted to estimates of parameters, $\hat{\theta}_k$, $k = 1, \dots, 3$, of the original model (equation 6) by regressing $\bar{\phi}_k$ on u for each treatment group via ordinary nonlinear least squares. The approximate standard error of $\hat{\theta}_k$ is

$$s(\hat{\theta}_k) \doteq \left[\left(\frac{\partial \phi}{\partial \theta'} \right)^{-1} \left(\frac{1}{n} S_{\phi} \right) \left(\frac{\partial \phi'}{\partial \theta} \right)^{-1} \right]_{kk}^{\frac{1}{2}},$$

and confidence intervals for θ_k were computed using the algorithm of section 2.3. Parameter estimates and approximate standard errors are given in table 4. For parameter A the point estimates obtained using the

expected value transformation, denoted by $\hat{\theta}_{\phi}^-$, are very similar to the standard two-stage estimates of table 1, but for the other two parameters the difference (as % of $\bar{\theta}$) can be large. The standard errors for $\bar{k}$ and $\bar{X}_0$ are larger than those for $\hat{k}_{\phi}^-$ and $\hat{X}_{0\phi}^-$ for all except the Mississippi chilled group. Note that in the expected value parameterization of the model we assume that the expected value parameters ϕ_1 , ϕ_2 , and ϕ_3 have homogeneous variances across treatment groups; whereas in the original parameterization we assume that the parameters A, k, and X_0 have homogeneous variances across treatments. The assumption that the expected values have homogeneous variances automatically leads to nonconstant variances for the original parameters when the parameter means differ among treatment groups. Also the confidence intervals based on the expected value transformation (fig. 3) for parameters k and X_0 are asymmetrical, in contrast to confidence intervals obtained by the standard two-stage method. The distributions of parameter estimates appear to be skewed for parameters k and X_0 , but symmetric for parameter A. This experiment provided little information on the distributions of the parameters k and X_0 for the Mississippi chilled group, and visual comparison of θ values among the four treatment groups only shows differences for parameter A, the maximum CO_2 uptake rate. The confidence intervals for parameters k and X_0 show clearly that point estimates are not adequate to characterize the responses of different treatments.

TABLE 4

Parameter estimates $\hat{\theta}_{\phi}^-$ and standard errors for RCR model
using expected value transformation

Group	Parameter Estimates			Standard Errors		
	A	k	X_0	A	k	X_0
QU Control	41.73	0.01083	53.28	1.798	0.001333	6.429
QU Chilled	39.08	0.009187	53.62	1.820	0.001079	6.805
MS Control	31.41	0.008886	46.10	1.826	0.001370	10.00
MS Chilled	17.80	0.01003	14.05	1.807	0.004044	35.30

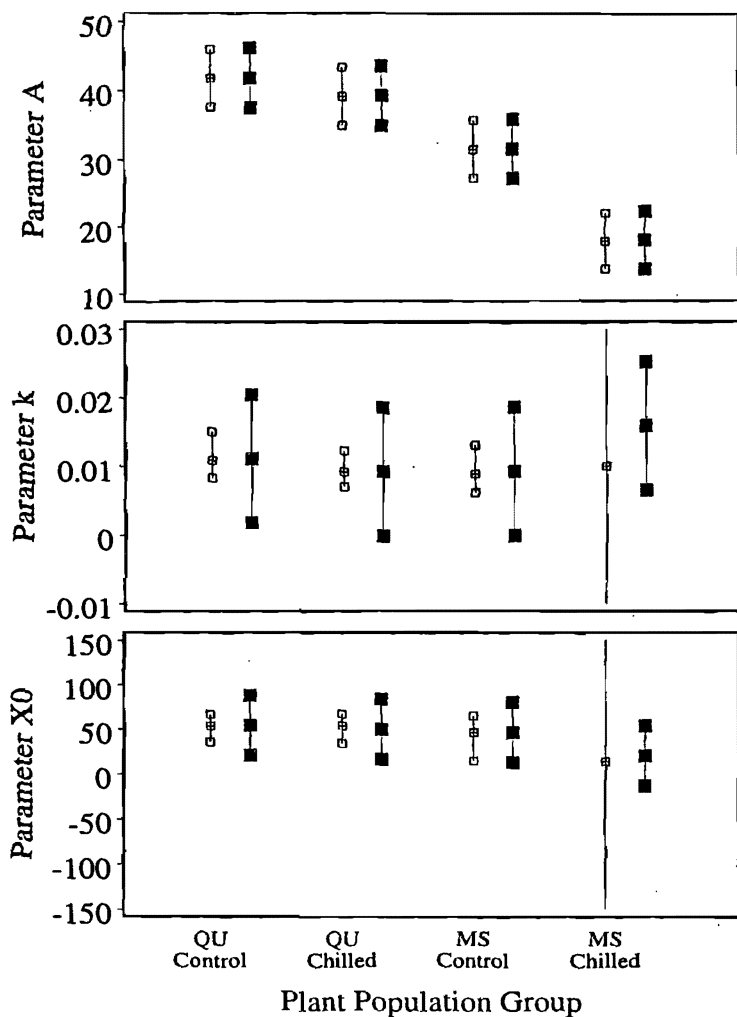


Figure 3. Confidence intervals for parameters of Mitscherlich function for CO_2 uptake for two plant populations and two chilling regimes. Open square: approximate profile likelihood method based on expected value transformation. Closed squares: standard two-stage method.

4. SIMULATION

A Monte Carlo simulation of an RCR Mitscherlich model was conducted to examine the coverage of confidence intervals for small samples under various levels of parameter effects curvature and under two different correlation structures for θ . In each Monte Carlo replication a sample of $n = 7$ vectors of $t = 5$ repeated measures observations was generated from one of the following distributions.

$$y_{ij} = A_i \left[1 - e^{-k_i(x_{ij} - X_{0i})} \right] + \epsilon_{ij}, \quad i = 1, \dots, 7; \quad j = 1, \dots, 5.$$

$$\theta_i = \begin{bmatrix} A_i \\ k_i \\ X_{0i} \end{bmatrix} \sim \text{MVN} \left[\begin{bmatrix} 1.0 \\ 0.01 \\ 0 \end{bmatrix}, \Sigma_{\theta k} \right], \quad \text{where } k = 1, 2 \text{ covariance patterns}$$

$$\Sigma_{\theta 1} = \begin{bmatrix} 0.01 & 0 & 0 \\ 0 & 0.000001 & 0 \\ 0 & 0 & 50 \end{bmatrix} \quad \text{or} \quad \Sigma_{\theta 2} = \begin{bmatrix} 0.01 & 0.00005 & 0.3536 \\ 0.00005 & 0.000001 & 0.003536 \\ 0.3536 & 0.003536 & 50 \end{bmatrix},$$

$\epsilon_{ij} \sim N(0, 0.001)$ iid, and ϵ_{ij} is independent of θ_i .

The two covariance patterns for θ correspond to: (1) no correlation among parameters A , k and X_0 ; and (2) correlation matrix $R = \begin{bmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 0.5 \\ 0.5 & 0.5 & 1 \end{bmatrix}$. We chose three sets of geometrically spaced design points: one that approximately covers the range $X_{0.50}$ to $X_{0.999}$, $\mathbf{x}1' = [50 \ 100 \ 200 \ 400 \ 800]$, and two that miss the low end of that range by varying amounts, $\mathbf{x}2' = [100 \ 200 \ 400 \ 800 \ 1600]$ and $\mathbf{x}3' = [150 \ 225 \ 337 \ 506 \ 760]$. It may be verified that curvature of the solution locus is much more severe for experiment 3 than for the other two. Two thousand Monte Carlo replications were run for each of the two covariance structures for the three within-individual experimental designs. For each Monte Carlo replication we computed the standard two-stage estimator $\bar{\theta} = \frac{1}{7} \sum_{i=1}^7 \hat{\theta}_i$ and the proposed estimator based on the

expected value transformation, $\hat{\theta}_{\phi}^- = f^{-1}(\bar{\phi})$. By thorough analysis in difficult cases, the least squares solutions were found in all simulations of experiments 1 and 2 and in 1966 simulations of experiment 3.

In order to illustrate the error rates in the two separate tails, table 5 illustrates the relative frequency with which the lower confidence limit exceeded θ and the relative frequency of the upper limit falling below θ . We would like the two tails of a confidence interval to have equal error rates so that the interval contains the values of the parameter with the highest likelihoods. This desirable state of affairs is achieved more closely using $\hat{\theta}_{\phi}^-$ than using $\bar{\theta}$.

TABLE 5

Simulation results. Coverage and asymmetry of 95% confidence intervals

Confidence Interval Error Rates (%)							
$\bar{\theta}$				$\hat{\theta}_{\phi}^-$			
	Parameter	lower [†]	upper	total	lower	upper	total
Design X1 m = 2000	A	2.7	2.0	4.7	2.6	2.2	4.8
	k	1.8	3.5	3.5	1.8	3.5	5.4
	X_0	3.4	2.0	5.4	2.5	2.9	5.4
Design X2 m = 2000	A	3.2	3.0	6.2	3.1	3.1	6.2
	k	2.0	3.0	5.0	2.6	3.0	5.6
	X_0	3.7	1.7	5.4	3.1	2.8	5.9
Design X3 m = 1966	A	3.4	1.4	4.8	2.4	1.8	4.4
	k	1.6	2.7	4.4	2.3	3.3	5.6
	X_0	2.8	1.1	3.9	2.3	2.8	5.0

[†] The headings $\bar{\theta}$ and $\hat{\theta}_{\phi}^-$ refer to limits calculated either by the standard two-stage method or by expected value transformations respectively. The entries in the column headed by "lower" are relative frequencies (%) of the lower limit exceeding true θ and "upper", correspondingly.

Table 6 gives the frequency distributions of the lengths of confidence intervals. Although the situation is somewhat confused by the undefined endpoints which sometimes occur with the profile likelihood limits, the results confirm the expectation that the $\hat{\theta}_{\phi}^{-}$ limits are narrower on average than the $\bar{\theta}$ limits.

TABLE 6

Frequency distributions of the lengths of 95% confidence intervals
from simulation experiments

Interval length	Design					
	X1		X2		X3	
	$\hat{\theta}_{\phi}^{-}\dagger$	$\bar{\theta}$	$\hat{\theta}_{\phi}^{-}$	$\bar{\theta}$	$\hat{\theta}_{\phi}^{-}$	$\bar{\theta}$
(i) Parameter A						
<0.100	119	120	107	106	90	91
0.100 - 0.200	1185	1189	1179	1187	1096	1148
0.200 - 0.300	648	645	672	677	619	677
>0.300	48	46	31	30	29	50
$\infty^{\dagger\dagger}$	0	0	11	0	132	0
(ii) Parameter K						
<0.003	735	696	175	159	18	20
0.003 - 0.006	1237	1252	1237	1074	452	346
0.006 - 0.009	28	50	516	548	751	566
0.009 - 0.012	0	1	52	120	408	396
>0.012	0	1	10	99	156	638
∞	0	0	10	0	181	0
(iii) Parameter X_0						
<16	324	345	3	3	0	0
16 - 40	1632	1578	455	436	13	9
40 - 160	44	77	1527	1521	1399	1167
>160	0	0	6	40	377	774
∞	0	0	9	0	177	16

$\dagger$ The column headed by $\hat{\theta}_{\phi}^{-}$ contains the frequencies following expected value transformations and that headed by $\bar{\theta}$ is untransformed.

$\dagger\dagger$ The classes designated by ∞ refer to confidence intervals of infinite length.

Although we do not show the sampling distributions here, $\hat{\theta}_\phi$ showed less bias than $\bar{\theta}$, especially in situations of greater curvature as in experiment 3.

5. DISCUSSION

For balanced designed experiments the standard two-stage method provides a simple method of estimating parameters and constructing confidence intervals for a nonlinear random coefficient regression model. Lindstrom and Bates (1990) comment that "...little is known about the properties of the 'naive' population estimators defined as simple weighted or unweighted means of nonlinear least squares individual estimates. The naive estimators will not be ML except in the case of completely balanced data with a linear expectation function. However, they are simple and intuitive and deserve further study". In this paper we have made such a study, from the perspective of trying to find an "optimal" set of parameter transformations whose estimators can be handled in a "naive" way.

The standard two-stage parameter estimates are biased, although the bias may be small for any particular design. The amount of bias may be estimated using Box's (1971) approximation (equation 2) to decide whether it is problematic. The bias and variance of parameter estimates are both reduced by using the expected value transformation before applying the standard two-stage method. In our notation, the estimators $\hat{\theta}_\phi$ are to be clearly preferred to $\bar{\theta}$ on the grounds of their sampling distributions.

A second important point concerns the confidence intervals for θ . One surprising result of this simulation study is that the coverage of the Wald-type confidence intervals from the standard two-stage method is excellent for all experimental designs considered. The shapes of the standard two-stage confidence intervals do not reflect the shapes of the distributions, however; and the standard two-stage confidence intervals are longer than the intervals based on the expected value transformation. More importantly,

the asymmetry of the intervals based on the expected value transformation avoids including values of θ in the confidence interval whose likelihood is lower than values outside the interval. Therefore, although both methods, in our simulation study, gave very similar coverage probabilities, the approximate profile likelihood method based on the expected value transformation is to be preferred to the standard Wald type limits.

One of the greatest advantages of the expected value transformation is that it focuses attention on the expected values rather than the parameters of the original model. Often in agricultural and environmental sciences researchers know what shape and properties a response curve ought to have, but not the specific mathematical formulation. This procedure brings the emphasis back to the basic data. It provides nearly unbiased estimates of the expected values, θ , and, in the case of the random coefficient regression model, leads us to use well established methods for multivariate normal data. It is then simple to perform such tasks as testing hypotheses comparing mean curves among groups or constructing confidence bands for fitted curves.

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